

PART-A

```
=====
GROUP 49: system.ruby.l1_cntrl0.L1Dcache STATS (7.0M-8.0M Cycles)
Window Hits      : 43542
Window Misses    : 6221
Miss Rate        : 0.125013
=====
```

The L1-D cache achieves a miss rate of approximately 12.5% between 7M–8M instructions, showing that most accesses are successfully handled at the first level. However, misses still occur and are served by lower memory levels, contributing to additional latency.

Part-b

For the DRAM

```
cse@cse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/gem5/m5out$ grep "simTicks" stats.txt
simTicks                                124640617500                # Number of ticks simulated (Tick)
cse@cse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/gem5/m5out$ echo "124640617500 / 500" | bc
249281235
cse@cse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/gem5/m5out$
```

For the NVM

```
cse@cse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/gem5$ cd ~/gem5/m5out
cse@cse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/gem5/m5out$ grep "simTicks" stats.txt
simTicks                                133494061000                # Number of ticks simulated (Tick)
cse@cse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/gem5/m5out$ echo "140032796500 / 500" | bc
280065593
cse@cse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/gem5/m5out$
```

DRAM Execution Cycles : 249281235

NVM Execution Cycles : 280065593

NVM takes more execution cycles than DRAM due to its higher read and write latencies. This results in slower memory access and increased overall execution time compared to DRAM.